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# An LLM-assisted Analog IC Design Tool with Automatic Topology Selection and Circuit Sizing

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***Abstract*—Recently, Large Language Models (LLMs) have gained a rising interest in different domains, including code generation for digital IC design. However, building an effective LLM for analog IC design is challenging because a valid analog circuit cannot be constructed by just randomly combining some codes. In this paper, an LLM-assisted analog IC design tool is presented to support from automatic topology selection to circuit sizing in a conversational manner. This tool allows designers to express their requirements through natural language rather than being restricted by a fixed template, which decreases the learning curve of using such an automation tool. In order to meet users' intentions, a subsequent ANN-based sizing process is invoked to further refine the "nearest" circuit obtained from the LLM without tedious iterations. Compared with previous works, the proposed LLM-assisted tool is more efficient and convenient while generating reliable analog designs to meet users' requirements. Our experiments show that the tool can generate workable circuits with short execution time.**

***Keywords—LLM, Zero-shot-CoT prompting, kNN search, automatic circuit sizing, ANN, circuit topology selection***

## I. INTRODUCTION

Because traditional analog IC design flow heavily relies on past knowledge and intuition, analog design automation has been studied for years to alleviate the long design cycles [1]. Most of existing works require a given circuit topology from designers, which still needs some prior knowledge to make a proper selection. A topology searching method in collaboration with a subsequent sizing process performed on a pre-built circuit library was proposed [4]. However, using decision tree traversal for topology selection often results in a long execution time when the number of candidates increases [4][14]. Further, [4] cannot take advantage of massive parallelism of graphics processing units (GPUs). Another approach uses artificial neural networks (ANNs) to merge topology selection and circuit sizing into one step, thus reducing the overall execution time [15]. Nevertheless, due to the uncertainty of ANN classification, inconsistent topology selection can occur under the same specification when users query at different time. This can be further exacerbated as the number of classification nodes increases. Therefore, it is important to find a good tradeoff between execution time and accuracy to be adopted by industrial designers.

Large Language Models (LLMs) have recently gained attention in Design Automation field because they allow IC designers to "discuss with an assistant" to construct a circuit. Thus, they can enable IC design tasks to become more easy and convenient without too much prior training. Despite a significant progress made in digital domain [2][13], the potential of LLMs in analog domain remains underdeveloped. Digital IC design automation can be thought of as a process of code generation and evaluation, safely ignoring effect of transistor sizes. However, due to the multifaceted constraints of analog ICs, building an effective LLM for analog IC design is more challenging because a valid analog circuit cannot be constructed by just randomly combining some codes. A recent work trains an LLM from both valid and invalid circuits to generate different topologies for analog designs with random performance [3]. In this approach, the randomly generated circuits are not guaranteed to be workable under the given specification, which is not acceptable for industrial designers. In practical applications, a reliable design tool is often desired. To utilize previous design experience, combining LLM with a trustworthy circuit library might be a good choice to help analog circuit design automation.

In this paper, we present an LLM-assisted analog IC (AIC) design tool that supports from automatic topology selection to circuit sizing in a conversational manner. Designers can express their requirements through natural language rather than being restricted by a fixed template, which decreases the learning curve of using such an automation tool. Based on an extendable circuit library, the proposed LLM-based topology selection is guided by Zero-shot-Chain-of-Thought (Zero-shot-CoT) prompting and k-nearest neighbors (kNN) search to achieve an extremely short execution time on getting a feasible solution from the trustworthy candidates. This approach is able to maintain reliable solutions at each usage iteration, which is more acceptable by analog designers. However, it is impossible to prepare all solutions in advance to meet any user's request. Therefore, to meet the user's intentions, a subsequent ANN-based sizing process is invoked to further refine the "nearest" circuit obtained from the library without tedious iterations. Compared with previous analog design automation approaches, the proposed LLM-assisted tool is more efficient and convenient while generating reliable analog designs to meet users' requirements.

## II. LLM-ASSISTED AIC DESIGN METHODOLOGY

### A. Workflow of the LLM-assisted AIC Design Tool

Fig. 1 illustrates the overall workflow of this tool. Starting from an input query, an LLM-based topology selection module finds a best-matched design from the given circuit library in terms of circuit performance. The topology and corresponding device sizes of this design will form the design context to answer the user's query. If the question requires some details of past conversations, the historical context is formed by retrieving relevant information from the chat log, which is then concatenated with the design context to form the total context for the final answer. This final answer will also be saved to the chat log for future use. Finally, if the best-matched design does not completely meet users' intentions, an ANN-based sizing help to refine the answer without tedious iterations. The following sections explain each component.

### B. Graphic User Interface (GUI)

Fig. 2 shows the GUI of the tool. There are 5 boxes for different purposes: inputting a user's query, receiving an answer, printing a best-matched design, displaying a netlist, and displaying a schematic. To perform different functions,

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there are 5 buttons in total: submitting a user's query, generating a best-matched design, activating the netlist and schematic display, performing a subsequent sizing, creating a simulation folder.

### C. Trustworthy Circuit Library

There are 12 circuit "feature names" in our trustworthy circuit library: gain, phase margin, unity frequency, bandwidth, dynamic power (with a 100kHz square wave input), static power, slew rate, estimated area, common mode rejection ratio (CMRR), power supply rejection ratio (PSRR), bias current, capacitance load. The library consists of 25 commonly used operational transconductance amplifiers (OTAs) and operational amplifiers (OPAMPs). Because each row in the library pertains to a vector including specifications of all feature names with corresponding device sizes and device values of a particular topology, the term "Specification Table" is used in Fig. 1.

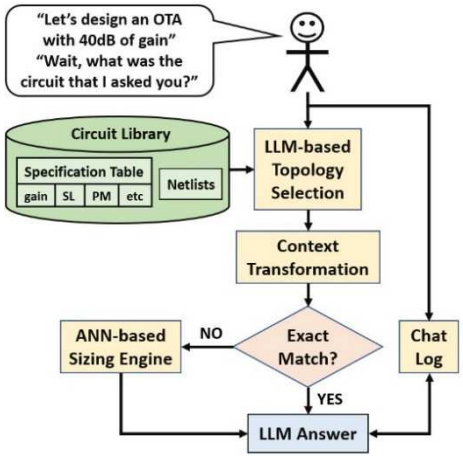



Fig. 1. The tool's workflow.

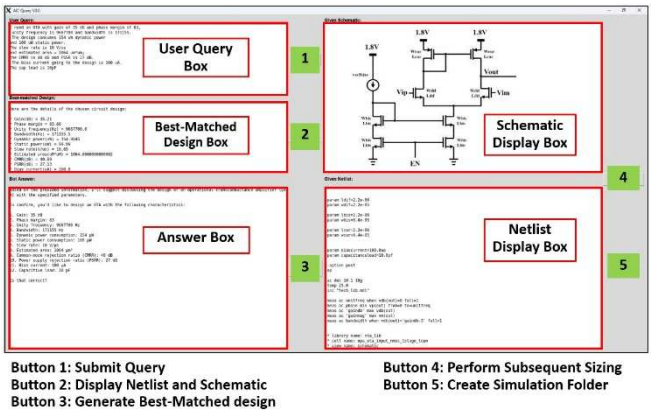



Fig. 2. The tool's GUI.

### D. Three-Stage Conversational Topology Selection

The topology selection module is conceptually divided into 3 stages. The module utilizes a general-purpose LLM, namely Llama 3.1 8B Instruct [5], in collaboration with our searching and prompting techniques, to quickly opt for a circuit topology in a conversational manner. Three stages are sequentially called "Summarization", "Retrieving k Circuits" and "Choosing Best-Matched", as shown in Fig. 3.

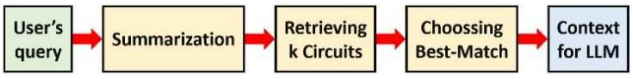



Fig. 3. Flowchart of the topology selection module.

In the "Summarization" stage, if T numbers of feature names are queried, a "Summarization" prompt automatically constructs T questions, each starts with a sentence "what is the required value of", followed by a feature name $f_c$. The sentence "You have to answer in bullet points" also needs to be present. The LLM takes the user's query and passes it through the "Summarization" prompt to generate a "Summary" in which the queried parameters are always arranged in the same order. This allows users to query about any circuit parameter in any order based on their thought processes. Furthermore, due to the LLM's capability to handle spelling mistakes, incorrect feature names can be mended so that they are consistent with the circuit library. Fig .4 illustrates the process of making a "Summary" from a query.

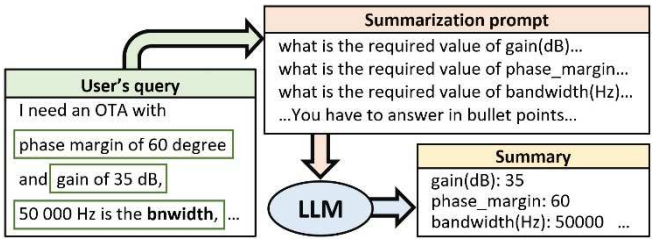



Fig. 4. The process of making a "Summary" from a query.

The "Summary" is fed to the "Retrieving k Circuits" stage. Firstly, a regular expression parses the "Summary" to obtain a "Performance Mapping Sequence" denoted by:

$$S_{perf} = \{v_1(f_1), v_2(f_2), \ldots, v_T(f_T)\} \quad (1)$$

where $v_1(f_1)$ is the first feature's value, $v_2(f_2)$ is the second feature's value, etc. The final feature's value is $v_T(f_T)$ if there are T queried feature names. Secondly, a kNN search [7] is performed on the library to retrieve k nearest data vectors of the $S_{perf}$. Empirically, we set hyperparameter k=4 and utilize Euclidean Distance Metric in our tool. Thirdly, an "Unrolling" procedure transforms the k vectors into k paragraphs by starting each paragraph with our customized phrase "design log", followed by a row index i, a ":" symbol, a feature name $f_c$, a "=" symbol, and a feature value $v_c(f_c)$. It makes a series of T descriptive pairs for each of the k resultant paragraphs. The kNN search is illustrated in Fig. 5. An example of the "Unrolling" procedure is illustrated in Fig. 6.

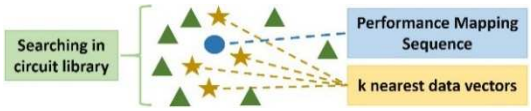



Fig. 5. kNN search illustration.

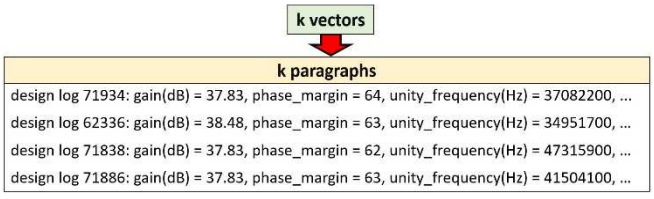



Fig. 6. An example of the "Unrolling" procedure.

In the "Choosing Best-Matched" stage, we employ Zero-shot-CoT prompting [6] by adding "Let's think step by step" in our "Best-Matched" prompt. In [6], experiments show that by adding the sentence, LLMs are forced to divide individual problems into several steps to provide individual answers. In our work, the specific problem is to compare each feature

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value $v_c(f_c)$ in each of the k paragraphs with each feature value in the $S_{perf}$ to opt for a best-matched design. As can be seen from Fig.7, the "Best-Matched" prompt is constructed by a concatenation of a fixed paragraph called "Request 1", the k circuit paragraphs, another fixed paragraph called "Request 2", and the original user's query. As illustrated in Fig. 7, The LLM in our tool takes the "Best-Matched" prompt as an input to produce the best-matched design, which is a circuit with device sizes that satisfies a query. This design is also the output of the topology selection module.

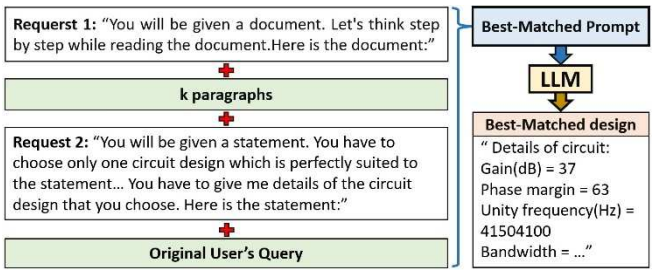



Fig. 7. Struture and functionality of the "Best-Matched" prompt.

The purpose of the kNN search is to return several designs with shortest distances from a $S_{perf}$. It is more practical than returning only one vector because there would be cases where the design that aligns with a user's intentions lies in a vector that is further away from a specific $S_{perf}$ than the closest vector. For example, a vector $V_1$={40dB,15kHz} is further away from a $S_{perf}$ ={38dB,17kHz} than a vector $V_2$={37dB,18kHz} but the query contains verbal conditions such as "gain>38dB". Thus, Zero-shot-CoT prompting can collaborate with kNN search to obtain a more aligned result in such case.

### E. Automatic Circuit Sizing

When a retrieved circuit does not meet the requirement, a subsequent sizing can be performed. ANN-based sizing is suitable in our case because of 2 reasons. Firstly, data from the library is reused in our training process. Secondly, because no new topology is added in the current version of the tool, sizing an existing circuit with a predetermined target from a query can take advantage of the extremely fast inference time of ANNs. We adopt the sizing method featuring ANNs from [8] with 2 major changes. In [8], error metrics are mean absolute error (MAE) and mean squared error (MSE). Both MSE and MAE decrease in the training process to approach zero without definitive ratios. Thus, R-Square Score (R2-score) [9] is an addition in our work to increase the interpretability of the results. As MSE decreases, R2-score approaches 1.0 (a ratio), meaning that the error between actual and predicted data is gradually minimized in training and evaluation. Another departure from [8] is that polynomial features are not utilized because we have not observed any improvement as compared to the case where only original data are used.

### F. Information Retrieval from Chat Log

A chat log is integrated so that users can query about the conversation history. To achieve the correct contexts from the chat log, instead of relying on dense retrieval [10], we employ a sparse retriever based on BM25 algorithm [11] for 2 reasons. Firstly, as the log becomes lengthier, creating a dense embedding vector of the whole log is significantly slower than calculating BM25 index. Secondly, sparse retrieval permits exact term matches which is important in searching for technical terms. This is pointed out in [12]. A hybrid retriever, a combination of a sparse retriever and a dense retriever, can also be discarded due to its slow runtime.

## III. Experimental Results

In Section III, the topology selection module takes $T_{kNN}$ to retrieve k circuits and takes $T_{BEST}$ to choose the best-matched design. Circuit areas are estimated without routing. In Section III, "N.T." stands for "Number of Topologies" and "I.S.M." stands for "Input Spelling Mistakes". Circuit simulations are performed by HSPICE with TSMC 0.18um process.

### A. Case study 1: Circuit Retrieval with Performance Match

In case study 1, the required OTA with existing performance and device sizes in the library is a simple 1-stage OTA (single-stage differential amplifier). Therefore, extra sizing is not required in this case, whose execution time can be denoted as $T_{EXE\text{-}CASE1} = T_{kNN} + T_{BEST}$. We repeat the experiment 5 times and record the longest $T_{EXE\text{-}CASE1}$ to compare with execution time in [4]. While [4] takes 11 min 27 sec to execute, our tool only requires 1 min 12 sec on a NVIDIA L4 GPU, as shown in Table II. The question-answering process of case study 1 is illustrated in Fig. 8 and Fig. 9. The word "phs margin" is corrected as "phase margin". The result of the retrieved circuit is summarized in Table I.

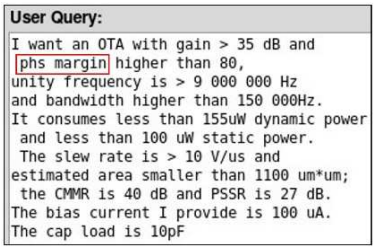



Fig. 8. Inputting a query into our tool in case study 1.

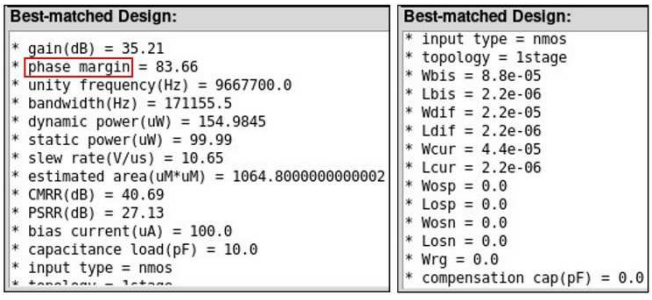



Fig. 9. Main parts of a Best-Matched Design from our tool in case study 1.

### B. Case study 2: Circuit Retrieval and Subsequent Sizing

In case study 2, the required circuit performance has no matching data in the library. The nearest sample is chosen by the kNN algorithm, whose performance does not meet the expected values. Therefore, a subsequent sizing is invoked that lasts for $T_{SIZE}$. The confirming simulation lasts for $T_{SIM}$. As a result, the execution time can be denoted as $T_{EXE\text{-}CASE2} = T_{kNN} + T_{BEST} + T_{SIZE} + T_{SIM}$.

TABLE I. Performance Result of the Retrieved 1-stage OTA in Case Study 1

| Para-meters | Gain $(dB)$ | Estimat-ed Area $(\mu m^2)$ | Phase Margin (degree) | Unity Frequency $(MHz)$ | Band-width $(kHz)$ | Dynamic Power $(\mu W)$ | Static Power $(\mu W)$ | Slew Rate $(V/\mu s)$ | CMRR $(dB)$ | PSRR $(dB)$ | Bias Current $(\mu A)$ | Capacitance Load $(pF)$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Values | 35.21 | 1064.80 | 83.66 | 9.66 | 171.15 | 154.98 | 99.99 | 10.65 | 40.69 | 27.13 | 100.00 | 10.00 |

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TABLE II. COMPARISON OF CASE STUDIES IN OUR WORK WITH [4]

| Method | Execution Time | N.T. | Tool Interface | I.S.M. |
|---|---|---|---|---|
| [4] | 11 min 27 sec | 24 | Fixed Template | N/A |
| Case Study 1 | 1 min 12 sec | 25 | Conversational | Allowed |
| Case Study 2 | 2 min 33 sec | 25 | Conversational | Allowed |

We repeat the experiment 5 times and record the longest $T_{EXE\text{-}CASE2}$ to compare with execution time in [4]. While [4] takes 11 min 27 sec to execute, our tool only requires 2 min 33 sec on a NVIDIA L4 GPU, as shown in Table II. The question-answering process of case study 2, that handles I.S.M. of our tool, is illustrated in Fig. 10 and Fig. 11.

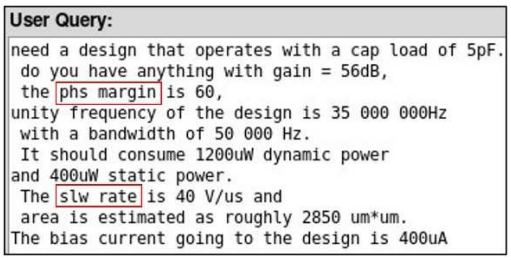



Fig. 10. Inputting a query into our tool in case study 2.

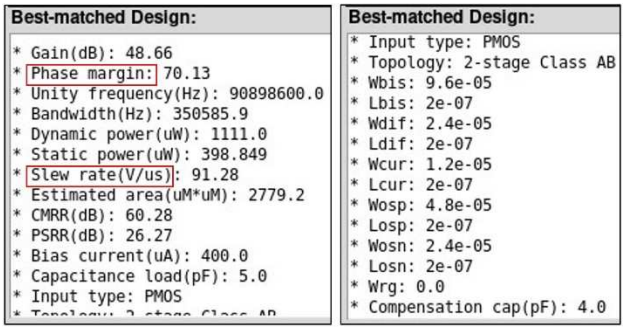



Fig. 11. Main parts of a Best-Matched Design from our tool in case study 2.

TABLE III. PERFORMANCE OF RETRIEVED CIRCUIT IN CASE STUDY 2

| Parameters | Expected | Best-Matched | After-Sized |
|---|---|---|---|
| Gain ($dB$) | 56.00 | 48.66 | 57.28 |
| Phase Margin | 60.00 | 70.13 | 61.24 |
| Unity Frequency ($MHz$) | 35.00 | 90.89 | 39.19 |
| Bandwidth ($kHz$) | 50.00 | 350.58 | 56.53 |
| Dynamic Power ($\mu W$) | 1200.00 | 1111.00 | 1104.40 |
| Static Power ($\mu W$) | 400.00 | 398.84 | 398.81 |
| Slew Rate ($V/\mu s$) | 40.00 | 91.28 | 68.34 |
| Bias Current ($\mu A$) | 400.00 | 400.00 | 400.00 |
| Capacitance Load ($pF$) | 5.00 | 5.00 | 5.00 |
| Estimated Area ($\mu m^2$) | 2850.00 | 2779.20 | 2841.86 |

The retrieved circuit, shown in Fig. 12, is a 2-stage class AB OPAMP. Table III illustrates the expected performance, best-matched performance and performance obtained after the circuit is sized. The after-sized performance become closer and slightly more optimized than the expected values. This demonstrates the capability of our tool to meet the intentions of users even if the design does not exist in the library.

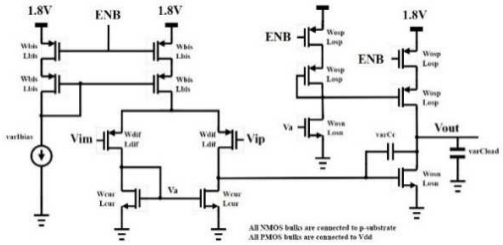



Fig. 12. The 2-stage classAB OPAMP displayed by the tool in case study 2.

## IV. CONCLUSION

In this paper, we present an LLM-assisted AIC design tool that achieves a competitive execution time to generate a workable circuit from a query. Future work would be to include an autoregressive circuit generation engine to enhance the capability of the tool.

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